

A Physics-Informed Neural Network-Based Scalable Model for GaN HEMTs

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Abstract—The integration of physical knowledge into artificial intelligence (AI)-based models for gallium nitride high electron mobility transistor (GaN HEMT) modeling is gaining increasing attention, as it reduces the amount of required training data, enhances the model’s physical plausibility, and contributes to the development of explainable AI. In this article, a physics-informed neural network (PINN)-based scalable GaN HEMT model is proposed. The model is trained with channel charge densities as fitting targets, while physical consistency is ensured by incorporating the drain current and gate capacitances into the PINN’s physical constraints. The model architecture and training algorithm are described in detail. The proposed PINN-based GaN HEMT modeling method achieves a high degree of automation in model development, while ensuring excellent prediction accuracy and physical plausibility. The significant self-heating and trapping effects in GaN HEMTs are also embedded into the proposed model. These features contribute to the model’s inherent and strong scalability. The developed GaN HEMT models of various sizes have undergone extensive small-signal and large-signal validation, demonstrating the model’s high accuracy and strong scalability.

Index Terms—Charge model, compact model, gallium nitride high electron mobility transistors (GaN HEMTs), large-signal modeling, physics-informed neural network (PINN).

I. INTRODUCTION

THE excellent radio frequency (RF) performance of gallium nitride high-electron-mobility transistors (GaN HEMTs) has made them highly valuable for applications in telecommunications, satellite communications, and defense industries [1]. Consequently, GaN HEMTs have received significant attention over the past decade. As a critical bridge between device and applications, device models have been attracting growing interest from both foundries and design houses. Due to the high-power and high-frequency characteristics of GaN HEMT devices, along with the presence of complex trapping [2] and thermal effects [3], [4], their modeling process faces significant challenges in terms of cost and

accuracy. Therefore, the development of a highly accurate and fully automated modeling methodology is of great importance for enabling the practical deployment of GaN HEMTs in a wide range of high-performance applications. In addition to conventional empirical [3], [4], [5], [6], [7], [8] and physical models [9], [10], [11], artificial intelligence (AI) techniques have been widely employed in GaN HEMT modeling over the past decade to enhance accuracy and enable automated model development. Previous studies have leveraged AI techniques to contribute to various aspects of GaN HEMT modeling, such as reducing development costs [12], [13], improving modeling accuracy [14], enhancing circuit design performance [15], [16].

In addition to these topics above, a pressing issue in the field of AI-based device modeling is how to effectively incorporate physical knowledge into purely mathematical neural network architectures to enhance their physical consistency and interpretability. To address this challenge, several investigations have been conducted. For instance, the knowledge-based neural network (KBNN) introduces physics-informed functions at the output layer of the neural network to improve the extrapolation capability of artificial neural network (ANN)-based device models [17], [18]. Recent work has also integrated ANNs into physical modeling frameworks to construct hybrid models, significantly reducing the development complexity of traditional physics-based models [19]. In addition, a space-mapping neural-network-augmented compact model is proposed to link device physics to model features for fast uncertainty quantification of GaN HEMTs under trapping effects [20]. Another study has incorporated subcircuits representing physically meaningful nonideal effects into ANN-based models to enhance their accuracy in GaN HEMT device modeling [21]. In addition, some studies have employed deep neural networks to facilitate parameter extraction for the physical ASM-HEMT model [22], [23].

In recent years, a novel neural network called the physics-informed neural network (PINN) [24] has attracted increasing attention across various fields. By incorporating physical laws as constraints of the neural network, this approach enables neural networks to achieve high prediction accuracy while maintaining excellent physical consistency and interpretability. PINNs also offer several other notable advantages, such as strong capabilities in solving both inverse and forward problems, as well as excellent adaptability to small datasets. Due to these strengths, PINNs have found broad applications in the field of electronic and microwave engineering, such

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as the modeling of microwave filter [25], the inverse design of electromagnetic periodic structures [26] and microwave waveguide [27], and real-time prediction of physical quantities of nanowire field-effect transistor [28].

In this work, we introduce PINN into the modeling of GaN HEMTs and propose a PINN-based scalable large-signal model. Benefiting from its strong physical interpretability, the proposed PINN-based model exhibits excellent scalability with respect to variations in device sizes and finger numbers. It also offers advantages such as a high degree of automation in model development and excellent accuracy. To validate the performance of the proposed model, comprehensive validation is conducted across devices of different sizes and finger numbers, including current–voltage (I – V), small-signal S-parameters, and large-signal nonlinear characteristics. Validation results demonstrate that the proposed model achieves modeling accuracy on par with mainstream modeling methods, while also exhibiting excellent scalability.

This article is organized as follows. Section II extracts the parameters required to construct the PINN-based model proposed in this article. Section III elaborates on the neural network architecture, training details, and modeling details of the proposed PINN-based GaN HEMT model. Section IV provides a comprehensive validation of the proposed model. Section V concludes this article.

II. MODEL PARAMETER EXTRACTION

This article focuses on 0.25- μm gate-length AlGaIn/GaN HEMTs fabricated using GaN-on-SiC technology. A cross-sectional schematic illustration of the device and its corresponding equivalent circuit are shown in Fig. 1. The parasitic elements consist of capacitances, inductances, and resistances, that is, C_{pg} , C_{pd} , L_g , L_d , L_s , R_g , R_d , and R_s . Specifically, C_{pg} and C_{pd} represent the crossover parasitic capacitances due to the coupling between the gate and source metals, and between the drain and source metals, respectively. L_g and L_d represent the inductance introduced by the gate and drain electrodes, while L_s accounts for the total inductance resulting from the source via holes and air bridges. R_g , R_d , and R_s represent the contact resistances of the gate, drain, and source metals, respectively. Encapsulated by these parasitic elements, the active region of the device consists of the intrinsic transistor, the source access region resistance, and the drain access region resistance, which are labeled $R_{acc,s}$ and $R_{acc,d}$ in Fig. 1(b). The source-side and drain-side 2-D electron gas (2DEG) densities of the intrinsic transistor, denoted as q_{is} and q_{id} , respectively, are the primary focus of this study and are highlighted in red in Fig. 1. These quantities are closely related to the gate capacitances and the drain current. For parameter extraction, the S-parameters were measured by sweeping the gate voltage from -4 to 1 V in steps of 0.25 V and the drain voltage from 0 to 40 V in steps of 2 V, with the frequency swept from 0.5 to 20.5 GHz in steps of 0.5 GHz.

After moving the reference planes of the S-parameters to the gate and drain manifolds of the device under test (DUT) [29], the parasitic elements as well as the source and drain access region resistances were further de-embedded [30], [31], [32]. As a result, the intrinsic GaN HEMT's Y-parameters can be

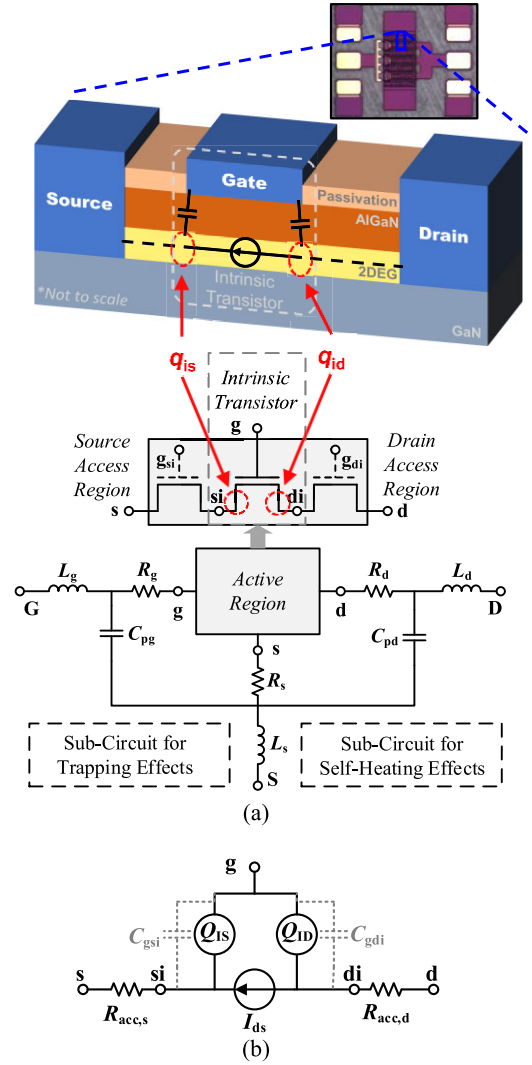


Fig. 1. (a) Cross-sectional schematic illustration of the investigated GaN HEMT, along with the corresponding equivalent circuit, and (b) large-signal equivalent circuit of the active region. The capacitances shown with dashed lines do not directly exist in the model but are derived from Q_{IS} and Q_{ID} .

determined. The nonlinear capacitance between the gate and the source-side of the intrinsic channel, denoted as C_{gsi} , can be expressed as

$$C_{gsi} = \frac{\text{Im}(Y_{11,int} + Y_{12,int})}{\omega}. \quad (1)$$

The capacitance between the gate and the drain-side of the intrinsic channel, denoted as C_{gdi} , can be expressed as

$$C_{gdi} = -\frac{\text{Im}(Y_{12,int})}{\omega}. \quad (2)$$

where ω represents the angular frequency.

The extracted C_{gsi} and C_{gdi} are plotted in Fig. 2. Then, they are converted into the equivalent source terminal charge (Q_{IS}) and drain terminal charge (Q_{ID}) by integrating with the intrinsic voltages V_{gsi} and V_{gdi} , as given by the following equations:

$$Q_{IS} = \int_{V_{gsi,min}}^{V_{gsi}} \Delta C_{gsi} dV_{gsi} \quad (3)$$

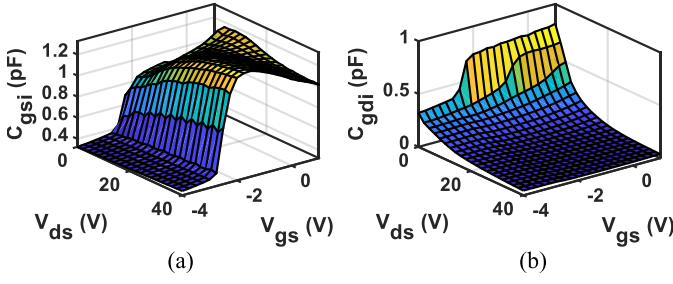


Fig. 2. Plots of the extracted intrinsic nonlinear (a) gate–source capacitance C_{gsi} and (b) gate–drain capacitance C_{gdi} .

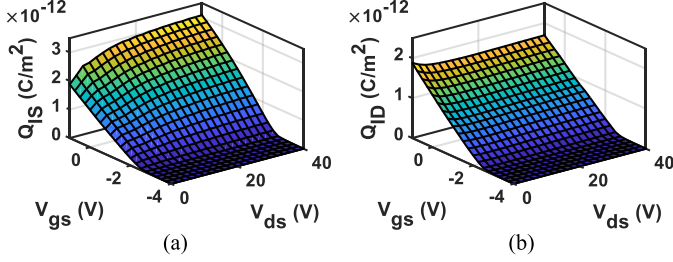


Fig. 3. Plots of the calculated (a) equivalent source terminal charge (Q_{IS}) and (b) equivalent drain terminal charge (Q_{ID}).

$$Q_{ID} = \int_{V_{gdi,min}}^{V_{gdi}} \Delta C_{gdi} dV_{gdi} \quad (4)$$

where ΔC_{gsi} and ΔC_{gdi} denote the difference from C_{gsi} and C_{gdi} between their minimum values $C_{gsi,min}$ and $C_{gdi,min}$ at OFF state. They are given by

$$\Delta C_{gsi} = C_{gsi}(V_{gsi}) - C_{gsi,min} \quad (5)$$

$$\Delta C_{gdi} = C_{gdi}(V_{gdi}) - C_{gdi,min}. \quad (6)$$

To compute the Q_{IS} and Q_{ID} at fixed intrinsic drain voltage, a set of uniformly spaced intrinsic drain-voltage values spanning the extracted V_{disi} range is introduced, which defines a series of fixed V_{disi} slices. For every such slice at a given V_{disi} , ΔC_{gsi} and ΔC_{gdi} are resampled using scattered-data interpolation, and trapezoidal integration is then performed along the resulting nonuniform V_{gsi} (or V_{gdi}) points. The results are shown as Fig. 3.

According to the Ward–Dutton charge partitioning scheme [33], the equivalent intrinsic charges Q_{IS} and Q_{ID} are quantitatively related to the source-side and drain-side charge densities (i.e., q_{is} and q_{id}), as given in the following equations:

$$Q_{IS} = \frac{2WLn_{gf}}{(q_{is}^2 - q_{id}^2)^2} \left(-q_{id}^2 \frac{q_{is}^3 - q_{id}^3}{3} + \frac{q_{is}^5 - q_{id}^5}{5} \right) \quad (7)$$

$$Q_{ID} = \frac{2WLn_{gf}}{(q_{is}^2 - q_{id}^2)^2} \left(q_{is}^2 \frac{q_{is}^3 - q_{id}^3}{3} - \frac{q_{is}^5 - q_{id}^5}{5} \right) \quad (8)$$

where W , L , and n_{gf} represent the gate width (single finger), gate length, and number of fingers, respectively.

After obtaining Q_{IS} and Q_{ID} , q_{is} and q_{id} are recovered by solving the corresponding inverse problem, which involves a high-order nonlinear system. This inversion is performed using the MATLAB fsolve numerical solver. For each bias

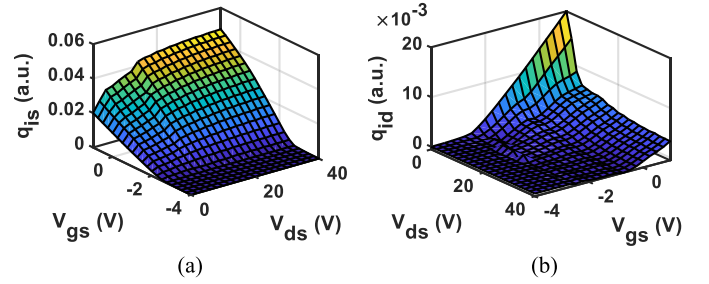


Fig. 4. Plots of the calculated (a) source-side 2DEG density (q_{is}) and (b) drain-side 2DEG density (q_{id}).

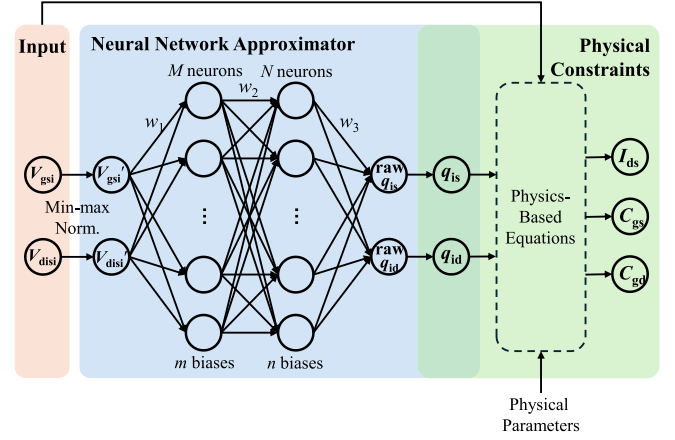


Fig. 5. Network architecture of the PINN-based GaN HEMT model.

point, fsolve is executed with an initial guess of $5e^{-3}$ for both q_{is} and q_{id} and with tight tolerances (FunctionTolerance, StepTolerance, and OptimalityTolerance set to $1e^{-16}$). The solved q_{is} and q_{id} are illustrated in Fig. 4. These charge densities are important physical quantities associated with both intrinsic gate capacitances and drain current I_{ds} [11], and they will be used to train PINN in Section III. It is worth noting that, in the following validations, the equivalent intrinsic charges calculated from q_{is} and q_{id} will be used directly in the simulator, and the intrinsic gate capacitances and drain current I_{ds} are derived from them [11], [34], [35].

III. PINN-BASED GAN HEMT MODEL

This section elaborates on the development of the PINN-based GaN HEMT model. This approach ensures that the developed model maintains excellent accuracy and physical plausibility, while significantly enhancing the level of automation in the model development process. Based on this method, the model development procedure will be introduced, including the construction and training of the PINN, the development of scalability, and the modeling of nonideal effects.

A. PINN Construction and Training

The PINN method is employed for GaN HEMT modeling, and its construction and training details are elaborated on in this section. The overall architecture of the PINN for GaN HEMT model is illustrated in Fig. 5, which consists of two

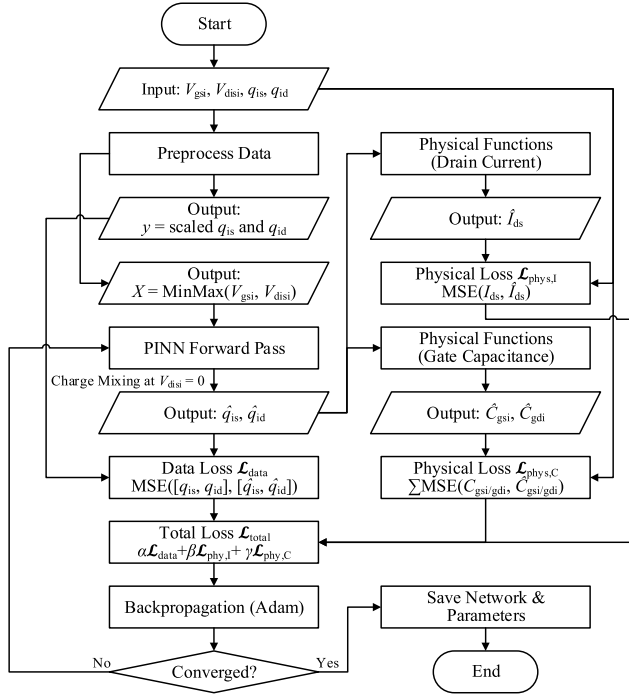


Fig. 6. Flowchart of the training process for the proposed PINN-based GaN HEMT model.

main components, that is, the neural network approximator and the physical constraints. The inputs to the PINN are intrinsic voltages V_{gsi} and V_{dsi} .

As the charge densities q_{is} and q_{id} are associated with both intrinsic gate capacitances C_{gsi} , C_{gdi} , and drain current I_{ds} , they are chosen as fitting targets. By fitting these charge densities, the device model aims to ensure both accurate RF and DC performance. In theory, given the accurate representations of q_{is} and q_{id} , both the calculated drain current and gate capacitances should accurately match their measured values. In practice, however, due to inaccuracies in the equivalent circuit and the q_{is} and q_{id} fitting, the I_{ds} , C_{gsi} , and C_{gdi} calculated from q_{is} and q_{id} models can hardly perfectly match the measured values simultaneously. To address this, the q_{is} and q_{id} models have to be fine-tuned to balance both DC and RF accuracy. This fine-tuning process is automated and constrained in the physical constraint module, where the modeled q_{is} and q_{id} are converted to I_{ds} , C_{gsi} , and C_{gdi} . These computed results are then compared with measurements, and the resulting errors are backpropagated to constrain the network training, thereby enabling high-accuracy fitting and reliable prediction of GaN HEMT characteristics. The flowchart of this procedure is outlined in Fig. 6, and a detailed technical explanation covering the neural network approximator and physical constraints is provided below.

In the neural network approximator part, the intrinsic voltages V_{gsi} and V_{dsi} , which serve as inputs to the PINN, are first normalized using min-max normalization. These normalized inputs are subsequently provided to the neural network for learning the nonlinear relationships between the input features and the target outputs, that is, q_{is} and q_{id} . A multilayer perceptron (MLP) is implemented in Tensorflow (Python 3.12),

and the Spyder integrated development environment (IDE) is used for model training. Two hidden layers are employed to simultaneously predict q_{is} and q_{id} , with each hidden layer consisting of six neurons. In this case, the total number of trainable parameters in the network is 74. This is determined primarily based on the size of the training set (794 samples) and the “one-in-ten” rule of thumb (Uncle Bernie’s rule) [36]. The hyperbolic tangent activation function is selected for the hidden layers, as it has strong nonlinearity while saturating for excessively large or small inputs. This choice enhances the robustness of the network when fitting nonlinear targets and improves reliability under extreme input conditions. For the output layer, the Sigmoid activation function is adopted to ensure sufficient nonlinearity while constraining all outputs to be strictly positive. A total of 90% of the solved q_{is} and q_{id} data, as shown in Fig. 4, is allocated for training, while the remaining 10% is reserved as an independent test set. A 10-fold cross-validation is conducted within the training data to monitor the model’s generalization throughout training without relying on an explicit validation set.

It is worth noting that when the voltage difference between the drain and source terminals of the intrinsic GaN HEMT is zero (i.e., $V_{dsi} = 0$), the charge densities inside the channel should theoretically be equal, meaning $q_{is} = q_{id}$. To ensure this condition is always satisfied, we first average the network’s raw predicted outputs for $q_{is} = q_{id}$ at $V_{dsi} = 0$ and obtain q_0

$$q_0 = \frac{\hat{q}_{is,raw}|_{V_{dsi}=0} + \hat{q}_{id,raw}|_{V_{dsi}=0}}{2}. \quad (9)$$

Then the final predicted outputs from the neural network are defined as follows:

$$\hat{q}_{is} = w(V_{dsi}) q_0 + (1 - w(V_{dsi})) \hat{q}_{is,raw} \quad (10)$$

$$\hat{q}_{id} = w(V_{dsi}) q_0 + (1 - w(V_{dsi})) \hat{q}_{id,raw} \quad (11)$$

where $w(V_{dsi})$ is a Gaussian-like smoothing weight

$$w(V_{dsi}) = \exp\left(-\frac{V_{dsi}^2}{2\sigma^2}\right). \quad (12)$$

Here σ is a small positive constant controlling the smoothing range. In this work, an appropriate σ ensures a smooth transition of the predicted q_{is} and q_{id} from the regime where V_{dsi} is near zero to regimes where V_{dsi} is significantly nonzero. Specifically, when V_{dsi} is 0, the weight w becomes to 1, enforcing that the predicted q_{is} equals q_{id} . Conversely, when V_{dsi} is significantly larger than σ , w becomes to 0, and the network’s raw predicted outputs are used directly. In this work, σ is set to 0.1 so that 99.9% consistency is ensured between the corrected and uncorrected q_{is} and q_{id} predictions when V_{dsi} is higher than 0.37 V. This setting also prevents excessive variations of the predicted q_{is} and q_{id} , which may degrade the convergence of the model. This configuration guarantees that all computational steps in the MLP remain differentiable, allowing gradients to propagate seamlessly through both branches during training.

Based on that, the parameters in the neural network approximator are optimized by minimizing the following data loss function ($\mathcal{L}_{\text{data}}$):

$$\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_{n=1}^N \left(\left(q_{\text{is}}^{(n)} - \hat{q}_{\text{is}}^{(n)} \right)^2 + \left(q_{\text{id}}^{(n)} - \hat{q}_{\text{id}}^{(n)} \right)^2 \right) \quad (13)$$

where N denotes the number of samples used for PINN training.

Second, in the physical constraint part, both DC and RF characteristics measured from the DUT are utilized to constrain the neural network training. For the DC constraint, the nonlinear drain current computed based on the known q_{is} and q_{id} using the MVS modeling theory [11], [32] is utilized. The predicted drain current is expressed as

$$\hat{I}_{\text{ds}} = W n_{\text{gf}} \frac{v}{2C_{\text{inv}} V_{\text{DSAT}}} \frac{\hat{q}_{\text{is}}^2 - \hat{q}_{\text{id}}^2}{\left(1 + \left(\frac{q_{\text{is}} - q_{\text{id}}}{C_{\text{g}} V_{\text{DSAT}}} \right)^{\beta} \right)^{1/\beta}} \quad (14)$$

where the saturation velocity v and the saturation voltage V_{DSAT} are defined as

$$V_{\text{DSAT}} = \frac{v_{\text{sat}} L}{\mu} (1 - F_{\text{f}}) + 2SS\phi_{\text{T}} F_{\text{f}} \quad (15)$$

$$v = v_{\text{sat}} (1 - F_{\text{f}}) + 2\phi_{\text{T}} \frac{\mu}{L} F_{\text{f}} \quad (16)$$

and the Fermi function F_{f} is given by

$$F_{\text{f}} = \frac{1}{\left(1 + \exp \left(\frac{\max(V_{\text{GS}}, V_{\text{GD}}) - (V_{\text{th}} - \alpha\phi_{\text{T}}/2)}{\alpha\phi_{\text{T}}} \right) \right)}. \quad (17)$$

The definition of DC physical constraint requires a set of physical parameters, including the gate width (W), number of fingers (n_{gf}), gate-to-channel capacitance per unit area (C_{g}), surface-potential transition factor during weak-to-strong inversion (α), linear-to-saturation transition factor (β), threshold voltage (V_{th}), thermal voltage (Φ_{T}), subthreshold swing (SS), channel length (L), and low-field mobility (μ). They can be obtained from known physical conditions, such as $I_{\text{ds}}-V_{\text{ds}}$ curves, $I_{\text{ds}}-V_{\text{gs}}$ curves, and ambient temperature.

Based on the calculated drain current and measured drain current, the physical constraint loss function for DC performance ($\mathcal{L}_{\text{phys,I}}$) can be defined as

$$\mathcal{L}_{\text{phys,I}} = \frac{1}{N} \sum_{n=1}^N \left(I_{\text{ds}}^{(n)} - \hat{I}_{\text{ds}}^{(n)} \right)^2. \quad (18)$$

Meanwhile, for the RF performance constraint, the intrinsic nonlinear gate-source and gate-drain capacitances, that is, C_{gsi} and C_{gdi} , are used. The computation of C_{gsi} and C_{gdi} corresponds to the inverse operation of the q_{is} and q_{id} extraction procedure described in Section II. It is worth noting that, in theory, q_{is} and q_{id} computed from C_{gsi} and C_{gdi} should allow for the exact reconstruction of the original C_{gsi} and C_{gdi} through the inverse operation. However, solving for q_{is} and q_{id} from the system of two-variable quintic equations for Q_{IS} and Q_{ID} may yield multiple solutions, potentially compromising their accuracy. Therefore, we also incorporate the physical constraints of C_{gsi} and C_{gdi} , which are calculated from the predicted q_{is} and q_{id} , into the loss function to ensure

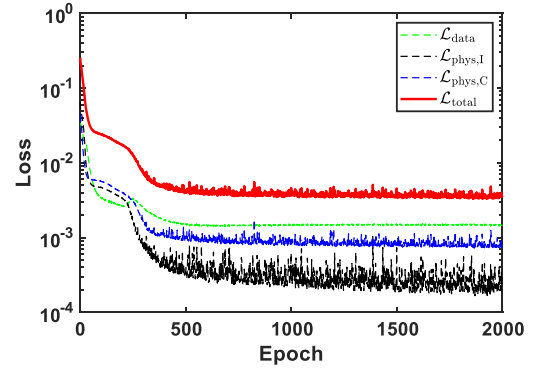


Fig. 7. Training loss curves of the PINN for GaN HEMT modeling over 2000 epochs.

the model's physical consistency. Consequently, the physical constraint loss function for RF performance ($\mathcal{L}_{\text{phys,C}}$) can be defined as follows:

$$\mathcal{L}_{\text{phys,C}} = \frac{1}{N} \sum_{n=1}^N \left(\left(C_{\text{gs}}^{(n)} - \hat{C}_{\text{gs}}^{(n)} \right)^2 + \left(C_{\text{gd}}^{(n)} - \hat{C}_{\text{gd}}^{(n)} \right)^2 \right). \quad (19)$$

Finally, the overall loss function, combining the neural network approximator and the physical constraints, is defined as follows:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{data}} + \beta \mathcal{L}_{\text{phys,I}} + \gamma \mathcal{L}_{\text{phys,C}} \quad (20)$$

where α , β , and γ are weighting coefficients to adjust the influence of each loss term. Since current and capacitance directly affect the model performance, the values for α , β , and γ are set as 1, 2, and 2, respectively. Due to the relatively small size of the dataset used in this study, the aforementioned loss-weighting configuration was determined through manual adjustment. With the expansion of the dataset in the future, it will be necessary to adopt adaptive loss weighting [37] to achieve faster and more stable training.

This loss function enables the training of q_{is} and q_{id} to simultaneously achieve optimal accuracy for both DC and RF model performance, while strictly preserving physical consistency throughout the training process. The network was trained using the Adam optimizer with a learning rate of 0.001. Training was conducted for a maximum of 2000 epochs without early stopping. As shown in Fig. 7, a rapid convergence of the loss function is observed, indicating that the design of the network architecture, hyperparameter configuration, and loss function in the proposed PINN is appropriate. Fig. 8 presents the fitting results for q_{is} and q_{id} , which exhibit slight differences from their initially extracted values. The corresponding RMSE values are 2.7×10^{-3} and 8.9×10^{-4} , respectively. However, this difference due to additional physical constraints in the physical losses leads to the optimal overall performance of the model in both DC (I_{ds}) and RF (C_{gs} and C_{gd}) regimes, as shown in Fig. 9. It should be noted that although the RMSE values between the optimized and initially extracted q_{is} and q_{id} appear to be not substantial, the adjustment of the initially extracted values is essential. Since q_{is} and q_{id} need to undergo a series of higher-power and differential operations to obtain I_{ds} , C_{gsi} , and C_{gdi} , relying solely on the initially extracted

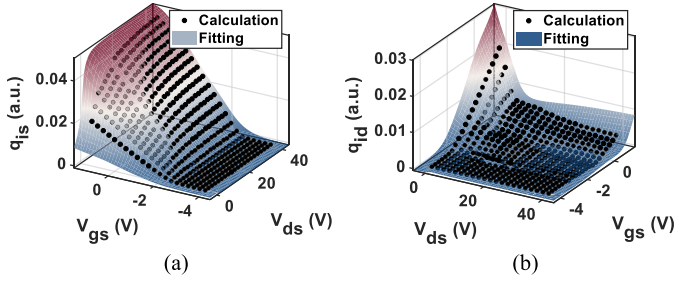


Fig. 8. Figures of the initially extracted (dots) and finally fit (surface) (a) source-side 2DEG density (q_{is}) and (b) drain-side 2DEG density (q_{id}).

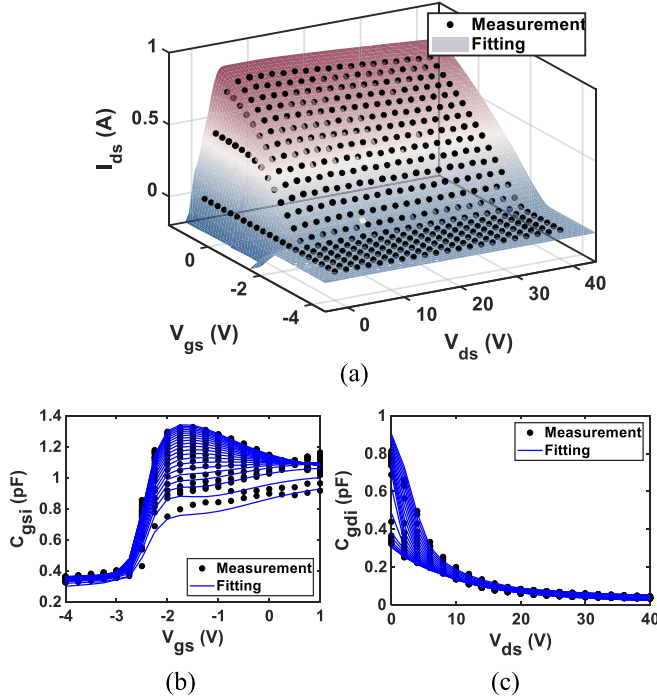


Fig. 9. Figures of the measured (dots) and fit (surface) (a) drain current (I_{ds}), (b) intrinsic gate-source capacitance (C_{gsi}), and (c) intrinsic gate-drain capacitance (C_{gdi}).

q_{is} and q_{id} without applying constraints from I_{ds} , C_{gsi} , and C_{gdi} can lead to significant problems. Even seemingly minor errors or abnormal fluctuations in the initial solutions of q_{is} and q_{id} can easily lead to physically unrealistic results, such as the disappearance of the knee voltage or the occurrence of negative capacitance. These issues ultimately make the model invalid for practical simulation. In this approach, we no longer tune the extracted values of q_{is} and q_{id} to achieve optimal I - V and S-parameter predictions. Instead, their predicted values are directly optimized through the training of the PINN, integrating the physical constraints into the learning process itself. In summary, this approach avoids physical inconsistency and extensive manual validation associated with the original purely ANN-based modeling method.

After training the PINN to fit q_{is} and q_{id} , the drain current I_{ds} can be obtained from q_{is} and q_{id} by recalculating the analytical expressions defined by the physical constraints, that is, (14)–(17). Similarly, the intrinsic gate capacitances C_{gsi}

and C_{gdi} can also be obtained by recalculating their equations in the physical constraints, which are exactly the inverse operation of the q_{is} and q_{id} extraction procedure. Since all these computations are analytical, they are highly compatible with common EDA simulation software.

B. Scalability

The scalability of the model is of significant importance to its applications. In this section, we propose a more refined scaling strategy by adjusting model parameters based on the previously established fixed-size model. This approach enables the development of a model that is scalable with respect to both the number of fingers and the gate length.

The extrinsic parameters of the model with different W_g and n_{gf} are modeled as follows:

$$C_{pg} = C_{pg0} + C_{pg1}W_g n_{gf} \quad (21)$$

$$C_{pd} = C_{pd0} + C_{pd1}W_g n_{gf} \quad (22)$$

$$L_g = L_{g0} + L_{g1}n_{gf} + L_{g2}W_g/n_{gf} \quad (23)$$

$$L_d = L_{d0} + L_{d1}n_{gf} + L_{d2}W_g/n_{gf} \quad (24)$$

$$L_s = L_{s0} + L_{s1}n_{gf} + L_{s2}W_g/(n_{gf}/2 + 1) \quad (25)$$

$$R_g = R_{g0} + R_{g1}W_g/n_{gf} \quad (26)$$

$$R_d = R_{d0} + R_{d1}/(W_g n_{gf}) \quad (27)$$

$$R_s = R_{s0} + R_{s1}/(W_g n_{gf}) + R_{s2}n_{gf}. \quad (28)$$

The coefficients of the equations above are listed in Table I. The equation of R_g differs slightly from that of R_d . In theory, contact resistance is inversely proportional to the contact area. However, since the gate metal is thin along the gate width direction, the gate width has a more pronounced effect on R_g . As a result, the R_g exhibits a positive dependence on the gate width [38]. To extract scaling parameters of C_{pg} , C_{pd} , R_g , and R_d in the above equations, at least two devices with different W_g and n_g are required. For L_g , L_d , L_s , and R_s , at least three devices with different W_g and n_{gf} are needed to determine the corresponding scaling parameters.

In the intrinsic part, it is worth mentioning that q_{is} and q_{id} represent the channel charge densities; therefore, they do not vary with the W_g and n_{gf} , which significantly lowers the difficulty for scalable model development. Benefiting from the use of physical equations, the values of I_{ds} , Q_{IS} , and Q_{ID} (i.e., I_{ds} , C_{gsi} , and C_{gdi}) for devices of different sizes can be directly calculated from their geometric parameters.

For the source and drain access region resistances, that is, $R_{acc,s}$ and $R_{acc,d}$, their values are inversely proportional to the W_g and n_{gf} , as shown below

$$R_{acc,s1}W_{g1}n_{gf1} = R_{acc,s0}W_{g0}n_{gf0} \quad (29)$$

$$R_{acc,d1}W_{g1}n_{gf1} = R_{acc,d0}W_{g0}n_{gf0} \quad (30)$$

where the subscript “0” refers to the parameters of the reference device, and “1” refers to those of the target device to which the scaling equation is applied.

C. Nonideal Effects

The model’s nonideal effects include self-heating and trapping, which are represented by the subcircuits shown in Fig. 10(a) and (b), respectively.

TABLE I
EXTRINSIC SCALING PARAMETERS

C_{pg0}	C_{pg1}	C_{pd0}	C_{pd1}	L_{g0}	L_{g1}	L_{g2}
0.035	5.2e-4	0.023	3.7e-4	0.001	1.182	0.683
L_{d0}	L_{d1}	L_{d2}	L_{s0}	L_{s1}	L_{s2}	
0	1.732	0.870	1.050	2.471	0.072	
R_{g0}	R_{g1}	R_{d0}	R_{d1}	R_{s0}	R_{s1}	R_{s2}
0.002	0.041	0.071	1.1e3	0	0.023	185.431

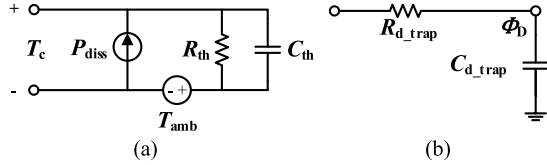


Fig. 10. Subcircuits to model (a) self-heating effects [3], [4] and (b) drain-voltage induced trapping effects [40].

Self-heating effects are modeled using the following equations [13]:

$$I_{ds,T} = I_{ds} (1 - K_T (T_c - T_{amb})) \quad (31)$$

$$K_T = \alpha_0 + \alpha_1 (1 + \tanh(\alpha_2 + \alpha_3 V_{ds})) \quad (32)$$

where α_0 – α_3 are fitting parameters. Channel temperature T_c is the sum of the ambient temperature T_{amb} and the self-generated heat. It is obtained using the subcircuit shown in Fig. 10(a) [3], [4]. In this work, the thermal resistance (R_{th}) in Fig. 10(a) is extracted using the maximum transconductance method [39].

To account for trapping effects, the equivalent gate voltage [4], [14] of the transistor is modified as follows:

$$V_{gs,eff} = \begin{cases} V_{gs} + (k_1 \tanh(V_{gs} - k_2) + k_3) (V_{ds} - \Phi_D) + k_4, & V_{ds} \leq V_{dsq} \\ V_{gs} + k_4, & V_{ds} > V_{dsq} \end{cases} \quad (33)$$

where k_1 – k_4 are fitting parameters that can be extracted under various quiescent voltage conditions. The dynamic parameter Φ_D is obtained from the equivalent circuit shown in Fig. 10(b) [40] to incorporate dynamic effects.

IV. DEVICE MODEL VALIDATIONS

In this section, the accuracy of the proposed scalable PINN-based model is validated through current–voltage (I – V), small-signal S-parameters, and large-signal measurements. The reference plane is calibrated at the drain and gate manifolds for S-parameter validation, and at the probe tips for large-signal validation. The device model with six fingers and 125- μ m gate width is first developed as the baseline model. Models with other gate widths and finger numbers are then scaled from this baseline model using the scaling methods discussed in Section II-B. We have also developed a classical ANN-based model and reproduced an ANN-physical model [19] as benchmarks to evaluate the performance of the proposed PINN-based model. Similar to the proposed PINN-based model, the ANNs employed in classical ANN-based

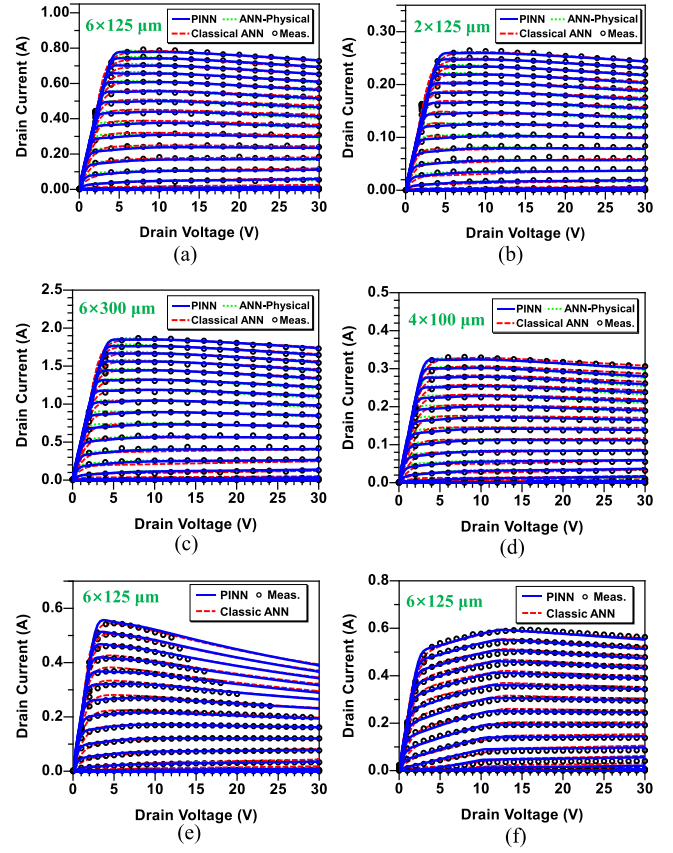


Fig. 11. Current–voltage characteristics comparison of (a) 6×125 - μ m, (b) 2×125 - μ m, (c) 6×300 - μ m, and (d) 4×100 - μ m GaN HEMT models developed with the proposed PINN method, the earlier proposed ANN-physical method [19], and classical ANN method. Both simulations and measurements are performed at $V_{dsq} = 0$ V, $V_{gsq} = 0$ V. Current–voltage characteristics comparison of the 6×125 - μ m GaN HEMT models developed with the proposed PINN method under (e) CW conditions and (f) pulsed conditions at $V_{gsq} = -3$ V, $V_{dsq} = 10$ V. Measurement setting (a)–(c) maximum gate voltage = 1 V with 0.25 V step and (d)–(f) maximum gate voltage = 0 V with 0.2 V step.

model and the previously proposed ANN-physical model share the same MLP architecture as used in this work (two hidden layers with six neurons per layer), to model the drain current and gate capacitances of devices with different dimensions, respectively. They are also trained using the Adam optimizer with a learning rate of 0.001 and a maximum of 2000 epochs, which are the same as those used in our proposed PINN-based model.

A. Current-Voltage and S-Parameter Validation

First, the pulsed I – V characteristics of the proposed PINN-based models are validated to confirm the accuracy of the presented I_{ds} model. In the pulsed I – V measurement, the pulse widths applied to the gate and drain are set to 0.6 and 0.5 μ s, respectively. The duty cycle is set to 0.01%, calculated based on the drain pulsewidth. Benefiting from the physical model architecture employed in this work, the PINN-based models exhibit excellent scalability. To verify the scaling accuracy, device models with different gate widths and finger numbers are developed and validated, as shown in Fig. 11(a)–(d). When

TABLE II
MODELING ERROR (NMSE) OF PULSED I - V CHARACTERISTICS

Device	PINN	Classical ANN	ANN-Physical
$6 \times 125 \mu\text{m}$ Fig. 11(a)	$7.8\text{E-}4$	$8.5\text{E-}4$	$6.7\text{E-}4$
$2 \times 125 \mu\text{m}$ Fig. 11(b)	$9.3\text{E-}4$	$9.9\text{E-}4$	$1.0\text{E-}3$
$6 \times 300 \mu\text{m}$ Fig. 11(c)	$5.2\text{E-}4$	$7.6\text{E-}4$	$4.6\text{E-}4$
$4 \times 100 \mu\text{m}$ Fig. 11(d)	$9.4\text{E-}4$	$1.0\text{E-}3$	$1.1\text{E-}3$

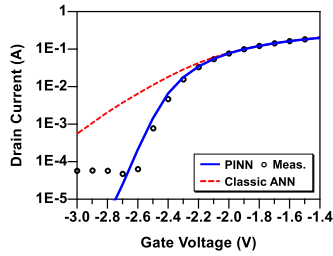


Fig. 12. Comparison of log-scale transfer curves between the proposed PINN-based model and the classical ANN-based model for the $6 \times 125 \mu\text{m}$ device. The drain voltage is fixed at 28 V.

validating the proposed PINN-based model, the gate width and finger number are used as external scaling parameters. Limited by the number of available device samples, the classical ANN-based model is developed separately for each gate width and finger number to ensure its performance. The results demonstrate that not only does the baseline PINN-based model ($6 \times 125 \mu\text{m}$) achieves high fitting accuracy, but other device models with different dimensions also exhibit excellent agreement with measured data. This indicates the good scalability accuracy of the resistive and current source components in the proposed PINN-based models. For better clarity, the I - V prediction errors, quantified by NMSE, for devices of different sizes are calculated and summarized in Table II. Following that, the self-heating and trapping effects are further validated, as shown in Fig. 11(e) and (f). It can be observed that the negative slope in the $I_{\text{ds}}-V_{\text{ds}}$ curves due to self-heating, as well as the threshold voltage shift caused by the trapping effects, are accurately captured by the proposed PINN-based models. The comparison indicates that the proposed PINN-based model has comparable overall $I_{\text{ds}}-V_{\text{ds}}$ accuracy to the classical ANN-based model and the previously proposed ANN-physical model. The advantages of the proposed PINN-based model are elaborated in detail in Section V.

Regarding the $I_{\text{ds}}-V_{\text{gs}}$ performance, it has been reported that classical ANN-based models have limited subthreshold and near-threshold accuracy [16]. However, since the proposed PINN-based model is constructed within a physics-based framework, it exhibits much better accuracy in the subthreshold and near-threshold regions, as illustrated in Fig. 12.

The small-signal S-parameter characteristics, which reflect the RF performance of the model, are further validated in this work. First, Fig. 13(a) and (b) shows that the $6 \times 125 \mu\text{m}$ device model exhibits excellent RF behavior in OFF and ON

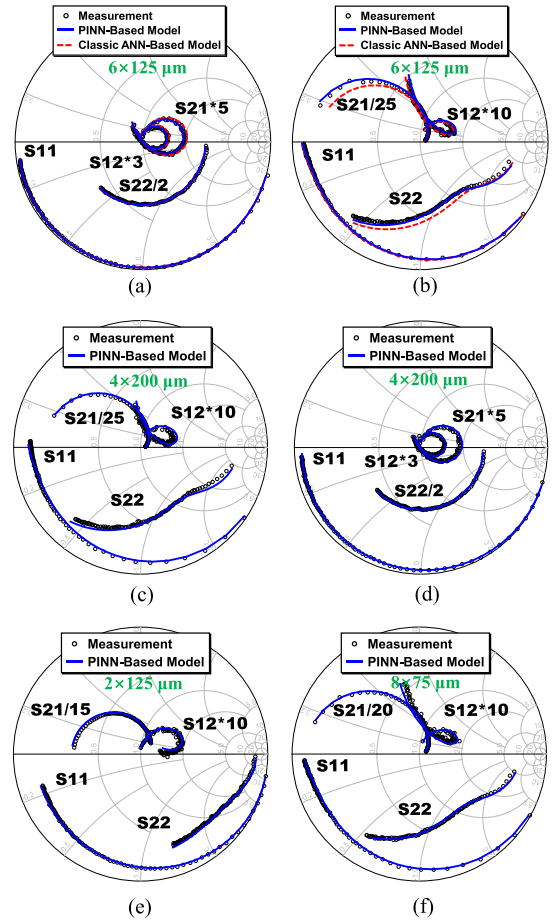


Fig. 13. S-parameter characteristics of the proposed PINN-based model and the classical ANN-based model for (a) $6 \times 125 \mu\text{m}$ device at $V_{\text{ds}} = 40 \text{ V}$ and $V_{\text{gs}} = -3.0 \text{ V}$, (b) $6 \times 125 \mu\text{m}$ device at $V_{\text{ds}} = 28 \text{ V}$ and $V_{\text{gs}} = -2.0 \text{ V}$; and of the proposed PINN-based model for (c) $4 \times 200 \mu\text{m}$ device at $V_{\text{ds}} = 28 \text{ V}$ and $V_{\text{gs}} = 0.0 \text{ V}$, (d) $4 \times 200 \mu\text{m}$ device at $V_{\text{ds}} = 40 \text{ V}$ and $V_{\text{gs}} = -3.0 \text{ V}$, (e) $2 \times 125 \mu\text{m}$ device at $V_{\text{ds}} = 28 \text{ V}$ and $V_{\text{gs}} = -2.0 \text{ V}$, and (f) $8 \times 75 \mu\text{m}$ device at $V_{\text{ds}} = 28 \text{ V}$ and $V_{\text{gs}} = -1.0 \text{ V}$. Frequency is swept from 0.5 to 20.5 GHz. Verification is performed under pulsed conditions.

states, respectively. Together with the high-accuracy pulsed I - V results in Fig. 11(a), these findings confirm that the proposed PINN-based model accurately captures both RF and I - V behaviors at the device level. Furthermore, Fig. 13(c) and (d) demonstrates that the scaled device models maintain high prediction accuracy for S-parameters, indicating that the proposed models exhibit excellent scalability for both capacitive and inductive components.

B. Large-Signal Validation

Large-signal validation not only allows a more comprehensive assessment of the accuracy of the model's nonlinear components but also better reflects practical requirements for RF power circuit applications. The large-signal measurements are not used in model training. Instead, the model is trained only with I_{ds} and small-signal data. The large-signal measurement setup is illustrated in Fig. 14. DC biases are applied to the DUT using the AM3200 pulsed I - V system, while a Maury automated mechanical tuner is employed to set the required source and load impedances. The Rohde and Schwarz

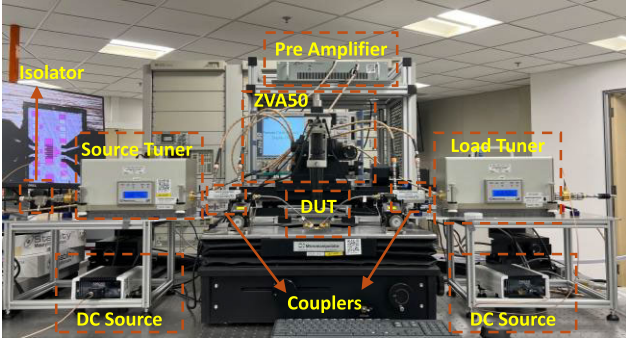


Fig. 14. Measurement setup for the large-signal validations of the proposed PINN-based GaN HEMT model.

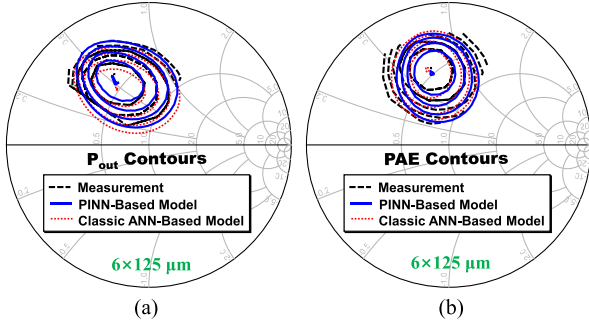


Fig. 15. Load-pull characteristics of the proposed $6 \times 125 \mu\text{m}$ PINN-based model and $6 \times 125 \mu\text{m}$ classical ANN-based model, showing (a) output power (P_{out}) contours under the load condition corresponding to the MxP and (b) PAE contours under the load condition corresponding to the MxE. Verification performed at 7 GHz, $V_{\text{ds}} = 28 \text{ V}$, and $V_{\text{gs}} = -2.0 \text{ V}$, with an input power of 22 dBm.

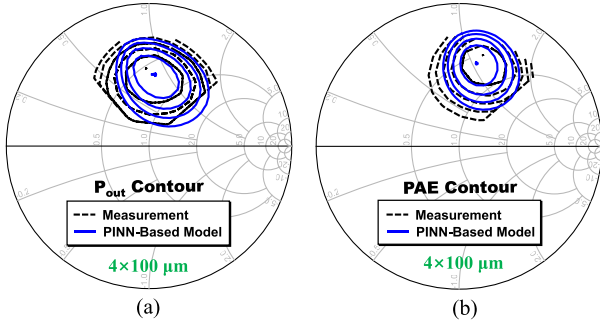


Fig. 16. Load-pull characteristics of the proposed $4 \times 100 \mu\text{m}$ PINN-based model, showing (a) P_{out} contours under the load condition corresponding to the MxP and (b) PAE contours under the load condition corresponding to the MxE. Verification performed at 9 GHz, $V_{\text{ds}} = 28 \text{ V}$, and $V_{\text{gs}} = -2.6 \text{ V}$, with an input power of 21 dBm.

ZVA50 vector network analyzer provides the input signal and measures both the incident and reflected waves at the input and output ports of the DUT via directional couplers. Due to the ZVA50's output power limitation of 5 dBm, a power amplifier with a saturated output power of 37 dBm and a maximum gain of 37 dB is placed between the ZVA50 and the source tuner to ensure sufficient input power. An isolator is placed between the power amplifier and the source tuner to protect the amplifier as well.

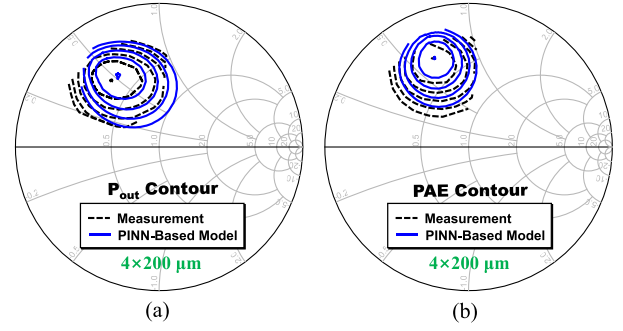


Fig. 17. Load-pull characteristics of the proposed $4 \times 200 \mu\text{m}$ PINN-based model, showing (a) P_{out} contours under the load condition corresponding to the MxP and (b) PAE contours under the load condition corresponding to the MxE. Verification performed at 9 GHz, $V_{\text{ds}} = 28 \text{ V}$, and $V_{\text{gs}} = -2.2 \text{ V}$, with an input power of 23 dBm.

TABLE III
MODELING ERROR OF LOAD-PULL CONTOURS

Device		Maximum Power		Maximum Efficiency	
		Value	Γ_L	Value	Γ_L
$6 \times 125 \mu\text{m}$	Mea.	36.5 dBm	$0.55 \angle 116^\circ$	58.5%	$0.55 \angle 103^\circ$
	(Fig. 15) Simu.	36.2 dBm	$0.51 \angle 118^\circ$	57.0%	$0.52 \angle 104^\circ$
$4 \times 100 \mu\text{m}$	Mea.	33.7 dBm	$0.55 \angle 92^\circ$	55.0%	$0.66 \angle 75^\circ$
	(Fig. 16) Simu.	33.6 dBm	$0.50 \angle 86^\circ$	54.1%	$0.60 \angle 79^\circ$
$4 \times 200 \mu\text{m}$	Mea.	35.9 dBm	$0.57 \angle 125^\circ$	55.8%	$0.67 \angle 112^\circ$
	(Fig. 17) Simu.	35.5 dBm	$0.58 \angle 120^\circ$	53.3%	$0.66 \angle 112^\circ$

Regarding the large-signal characteristics, the load-pull performance of the model was first validated. Each set of load-pull validations includes the contours corresponding to the device operating under maximum output power (MxP) and maximum power-added efficiency (MxE) conditions. For the baseline device with a gate finger configuration of $6 \times 125 \mu\text{m}$, the validation results for MxP and MxE load-pull contours are shown in Fig. 15(a) and (b), respectively. The corresponding results for scaled devices with $4 \times 100 \mu\text{m}$ and $4 \times 200 \mu\text{m}$ configurations are presented in Figs. 16 and 17, respectively. To provide a clear comparison of the load-pull performance, the key metrics extracted from Figs. 15–17 are summarized in Table III. As shown in Table III, the proposed PINN-based models accurately predict both the MxP and the maximum power-added-efficiency (PAE) values, as well as their corresponding load reflection coefficients.

Subsequently, the proposed models were evaluated under both MxP and MxE load conditions to validate their power sweep characteristics, specifically focusing on the trends of output power (P_{out}), transducer power gain, and PAE as input power (P_{in}) varies. Figs. 18 and 19 show that the baseline $6 \times 125 \mu\text{m}$ device model exhibits good agreement between measurement and simulation under the MxP and MxE conditions, respectively. The corresponding power sweep results for several scaled models are presented in Figs. 20–22. These results further confirm the high accuracy of the proposed PINN-based modeling approach in capturing the nonlinear characteristics of the device.

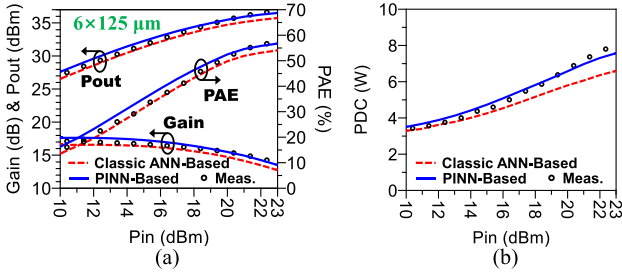


Fig. 18. (a) Large-signal power sweep and (b) P_{DC} characteristics of the proposed $6 \times 125 \mu\text{m}$ PINN-based model and the $6 \times 125 \mu\text{m}$ classical ANN-based model under the load conditions corresponding to MxP. Verification performed at 7 GHz, $V_{ds} = 28 \text{ V}$, and $V_{gs} = -2.0 \text{ V}$.

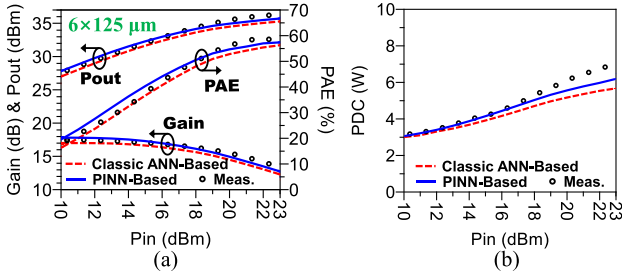


Fig. 19. (a) Large-signal power sweep and (b) P_{DC} characteristics of the proposed $6 \times 125 \mu\text{m}$ PINN-based model and the $6 \times 125 \mu\text{m}$ classical ANN-based model under the load conditions corresponding to MxE. Verification performed at 7 GHz, $V_{ds} = 28 \text{ V}$, and $V_{gs} = -2.0 \text{ V}$.

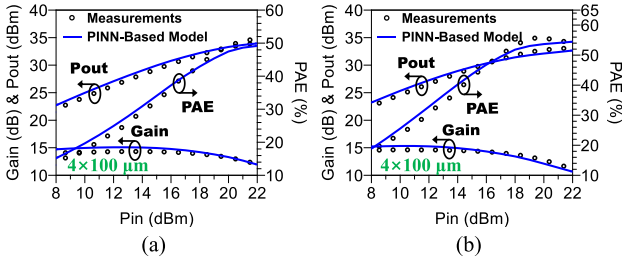


Fig. 20. Large-signal power sweep characteristics of the proposed $4 \times 100 \mu\text{m}$ PINN-based model under the load conditions corresponding to (a) MxP, and (b) MxE. Verification performed at 9 GHz, $V_{ds} = 28 \text{ V}$, and $V_{gs} = -2.6 \text{ V}$.

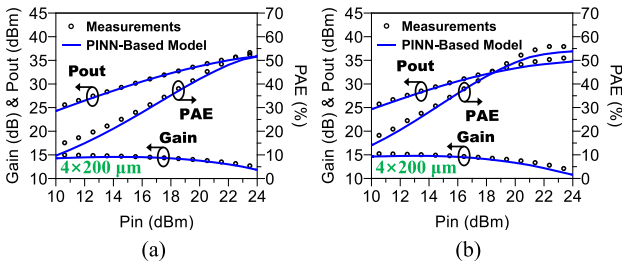


Fig. 21. Large-signal power sweep characteristics of the proposed $4 \times 200 \mu\text{m}$ PINN-based model under the load conditions corresponding to (a) MxP and (b) MxE. Verification performed at 9 GHz, $V_{ds} = 28 \text{ V}$ and $V_{gs} = -2.2 \text{ V}$.

Finally, the performance of the proposed PINN-based model was validated under different harmonic conditions. Specifically, power sweep characteristics at both the fundamental

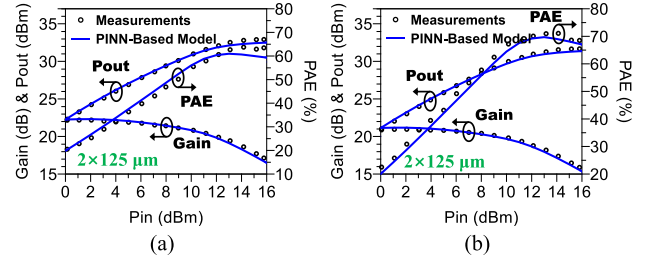


Fig. 22. Large-signal power sweep characteristics of the proposed $2 \times 125 \mu\text{m}$ PINN-based model under the load conditions corresponding to (a) MxP and (b) MxE. Verification performed at 5.8 GHz, $V_{ds} = 28 \text{ V}$, and $V_{gs} = -2.1 \text{ V}$.

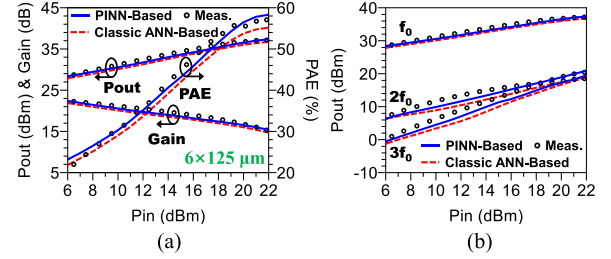


Fig. 23. Large-signal power sweep verification of the proposed $6 \times 125 \mu\text{m}$ PINN-based model and classical ANN-based model, showing (a) large-signal performance at the fundamental frequency (4 GHz) and (b) output power at various harmonics under load conditions corresponding to the MxP. Verification was conducted at $V_{ds} = 28 \text{ V}$ and $V_{gs} = -2.0 \text{ V}$.

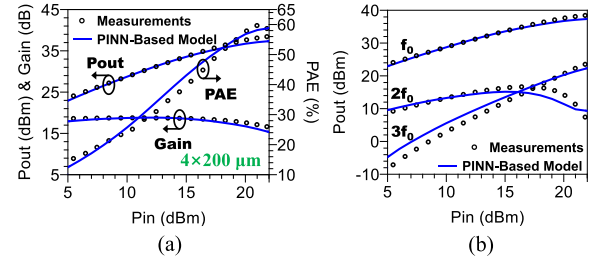


Fig. 24. Large-signal power sweep verification of the proposed $4 \times 200 \mu\text{m}$ PINN-based model, showing (a) large-signal performance at the fundamental frequency (4 GHz) and (b) output power at various harmonics under load conditions corresponding to the MxP. Verification was conducted at $V_{ds} = 28 \text{ V}$ and $V_{gs} = -2.5 \text{ V}$.

frequency and harmonic frequencies were evaluated for the $6 \times 125 \mu\text{m}$ and $4 \times 200 \mu\text{m}$ device models, as shown in Figs. 23 and 24, respectively. As illustrated in Figs. 23(b) and 24(b), the proposed PINN-based models accurately capture the harmonic distortion behavior of the devices.

In the power sweep results presented above, the maximum deviation of PAE from the predicted value is approximately 4%. Although this level of error is generally considered acceptable, it is essential to identify its underlying causes. In this work, we attribute this deviation primarily to the modeling of thermal and trapping effects. For GaN HEMT devices, the modeling of these effects significantly affects both the magnitude and time-domain waveform of I_{ds} . Consequently, this influences the prediction of IDD under large-signal conditions, which in turn degrades the accuracy of the key parameter P_{DC} in the PAE calculation.

As shown in the above comparisons between the proposed PINN-based model and classical ANN-based model, it can be observed that the proposed model achieves overall accuracy comparable to, or even slightly better than, that of the classical ANN-based model. Nevertheless, compared with the comparable or slightly improved accuracy, we believe that the more significant contribution of this work lies in using physical properties to constrain neural network training, thereby yielding a physically reliable and interpretable machine learning-based model for GaN HEMTs.

Additionally, extrapolation capability is another key aspect of the proposed PINN-based model. It is worth noting that the large-signal measurements are not used for training the model. Under high-power operation, the instantaneous voltage swings can exceed the bias ranges covered by the training data, and the good agreement in large-signal validation (Figs. 18 and 19) therefore provides evidence of bias extrapolation of the proposed model. Furthermore, the model is trained only on the $6 \times 125 \mu\text{m}$ device, while the measurements of other geometries are reserved for validation. The accurate predictions on these unseen geometries (I - V : Fig. 11; S-parameters: Fig. 13; large-signal: Figs. 16–24) further support geometry extrapolation.

In summary, this section comprehensively validates the proposed PINN-based scalable models, including I - V , S-parameters, and large-signal performance. The validation results confirm that the proposed PINN-based approach for GaN HEMTs achieves modeling accuracy comparable to mainstream ANN-based methods, while also demonstrating excellent scalability.

V. DISCUSSION

In this section, a comprehensive discussion of the proposed model is provided. We further discuss the necessity of using PINN in this work and provide a comprehensive benchmarking against other existing models to better highlight its distinctions.

A. Necessity of Using PINN

To clearly demonstrate the necessity of using PINN in this work, this section compares the model using the proposed PINN-based q_{is} and q_{id} with the model in which ANN is used to directly fit the initially extracted q_{is} and q_{id} . It should be noted that the initial extraction results of q_{is} and q_{id} contain noticeable noise and fluctuations, since their extraction requires solving a system of two fifth-order polynomial equations. Therefore, when training ANN directly on these initially extracted q_{is} and q_{id} , additional measures are applied to mitigate the impact of noise, including the use of L1 regularization and reducing the learning rate to 0.005.

Fig. 25 shows the I - V characteristics of the two models. The model without PINN fails to accurately predict the static operating point in the saturation region, which is the primary region of interest in this study. In the RF performance comparison shown in Fig. 26, this inaccuracy further leads to abnormal behavior in the magnitude of S21 and in the real part of S22 [Fig. 26(a)], as well as inaccurate large-signal performance [Fig. 26(b)]. Overall, these results indicate that

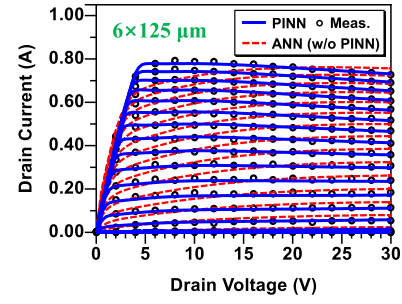


Fig. 25. Current-voltage characteristics of the $6 \times 125 \mu\text{m}$ device for models with PINN-based q_{is} and q_{id} and without PINN-based q_{is} and q_{id} (ANN direct fitting). The comparison is performed under pulsed conditions at $V_{dsq} = 28 \text{ V}$ and $V_{gsq} = -3.0 \text{ V}$.

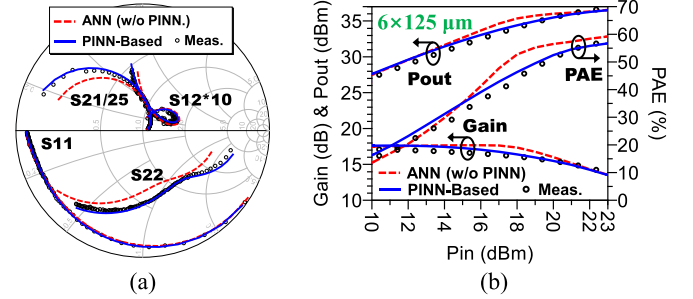


Fig. 26. (a) S-parameter and (b) power-sweep characteristics of the $6 \times 125 \mu\text{m}$ device for models with PINN-based q_{is} and q_{id} and without PINN-based q_{is} and q_{id} (ANN direct fitting). The S-parameters are performed at $V_{ds} = 28 \text{ V}$ and $V_{gs} = -2.0 \text{ V}$. The power-sweep are performed at maximum power condition, 7 GHz , $V_{ds} = 28 \text{ V}$, and $V_{gs} = -2.0 \text{ V}$.

models relying solely on the initially extracted q_{is} and q_{id} exhibit unsatisfactory prediction accuracy, demonstrating the necessity of using PINN in the proposed approach.

Based on the comparison above, it can be concluded that the PINN method introduces a moderate refinement to q_{is} and q_{id} . However, since q_{is} and q_{id} can affect the model through a series of higher-order terms and differential operations, even such moderate refinements can lead to significant improvements at the model level. The use of PINN becomes particularly meaningful under practical measurement conditions, where noise and systematic errors in the extraction of q_{is} and q_{id} are difficult to avoid. In this case, the physical constraints enforced in the PINN learning process help overcome the limitations when ANN is trained directly on the initially extracted q_{is} and q_{id} .

B. Comparison With Prior Work

The comparison in the previous sections has shown that the proposed PINN-based model achieves accuracy comparable to, or slightly better than, prior work. To further clarify the contributions of this work, we comprehensively compare the proposed PINN-based GaN HEMT model with two representative prior approaches: the classical ANN-based model and the previously proposed ANN-physical method [19], in addition to accuracy.

In brief, the classical ANN-based model exhibits a certain level of automation but lacks clear physical meaning; the

TABLE IV
COMPARISON OF THE PROPOSED PINN-BASED MODEL AND PRIOR WORK

Metrics	PINN-Based (This Work)	Classical ANN-Based	ANN-Physical [19]
Physical consistency with real devices	High	Low	High
Near-threshold accuracy	High	Low	High
Dataset optimization dependency	Not required	Not required	Highly dependent
Minimum manual inspection count per run	1	3	2
Error source	Model training	Model training	Dataset optimization + model training
Error source clarity	Clear (separated sources)	Clear (separated sources)	Unclear (entangled sources)
Degree of automation	High	Medium	Low
Performance consistency	High	High	Low
Total iterations performed	4	9	69
Actual workflow time (with the data already available)	Approximately 1.5 hours	Approximately 2.5 hours	Approximately 35 hours

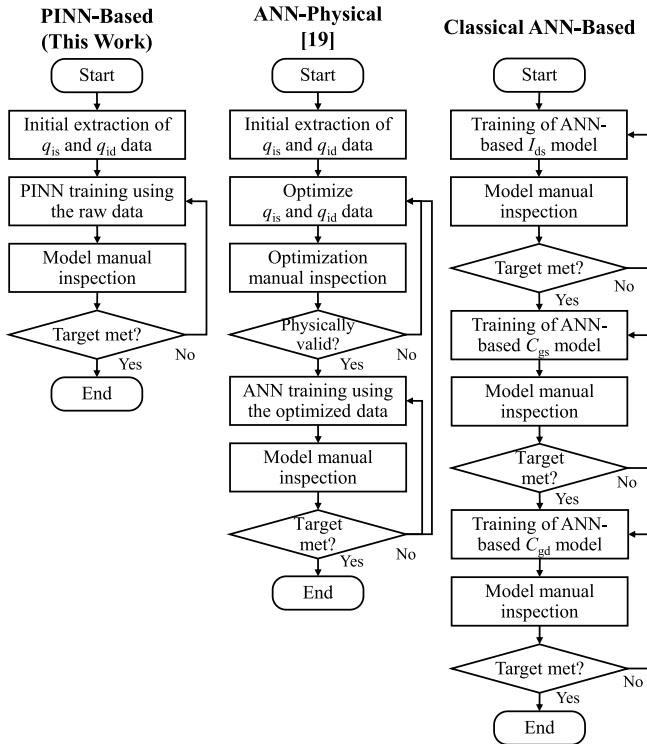


Fig. 27. Workflow comparison of the PINN-based model (this work), classical ANN-based model and hybrid ANN-physical model [19].

previously proposed ANN-physical method has clear physical meaning but suffers from limited automation and low performance consistency; by contrast, the proposed PINN-based model combines clear physical meaning with a high degree of automation and stable modeling performance.

These performance differences are further explained by the workflow analysis shown in Fig. 27. A posttraining manual inspection step is often necessary in machine-learning-based GaN HEMT modeling to assess the model performance, including its extrapolation capability and whether overfitting happens. In the proposed method, only a single manual inspection is required after the PINN training. In the classical

ANN-based model, separate ANNs are trained for the current source and the gate capacitances. Therefore, three manual inspections are needed per run, and the overall model lacks explicit physical constraints. In the previously proposed ANN-physical method, although physical constraints are introduced and the number of manual inspections per run is reduced to two, the reliance on dataset optimization makes the error sources ambiguous. For example, when discrepancies are observed between ANN predictions and the measured device behavior, it is difficult to attribute them clearly to the dataset optimization stage, to the ANN training process, or to both. Moreover, the dataset optimization also introduces additional uncertainty into the subsequent ANN training process, as it only ensures a minimized overall error and does not guarantee the performance of the optimized dataset in the subsequent ANN training. This further increases the uncertainty of model development. For example, the optimization algorithm cannot ensure the continuity of the optimized dataset. Substantial manual effort is therefore required to verify the optimization results. These factors reduce the degree of automation of the workflow in the previously proposed ANN-physical method and lead to higher run-to-run variability.

A comprehensive comparison of their characteristics is summarized in Table IV. The table not only summarizes the performance comparisons discussed above but also records their practical behavior during actual model development in the last two rows. For the proposed model, a total of four iterations (as illustrated in Fig. 27) are performed to achieve the development target, resulting in an actual development time of approximately 1.5 h. For the classical ANN-based model, to reach the same target, the training of I_{ds} , C_{gs} , and C_{gd} requires a total of nine iterations, with an actual development time of approximately 2.5 h. For the ANN-physical model proposed in the previously proposed ANN-physical method, due to the increased uncertainty introduced by dataset optimization and the substantial computational cost of the optimization process itself, a total of 69 optimization iterations are performed to achieve the development target, resulting in an actual development time of approximately 35 h. Therefore, these results further confirm that the proposed PINN-based GaN HEMT model achieves a higher degree of automation.

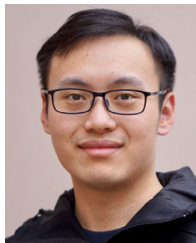
VI. CONCLUSION

A scalable large-signal model for GaN HEMTs based on PINN is proposed in this article. In the proposed approach, PINNs are employed to model charge densities of GaN HEMTs, which are closely associated with the device's DC and RF performance. The proposed approach offers several advantages, including a high degree of automation in model development, excellent modeling accuracy, and strong physical plausibility. This inherent physical interpretability further contributes to the model's superior scalability. Extensive validation across devices of different dimensions further verifies its high accuracy and excellent scalability.

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